

SM-1 Service module

Technical specifications and assembly instructions



Description

The SM-1 is a low tech, computerless and recoverable service module. The SM-1 is designed to operate with naked capsules on top and to provide both launch services and orbital maneuvering capabilities.

Recovery is done with parachutes. The stage is able to re-enter Kerbin's atmosphere on its own, requiring only a new set of fins during refurbishment.

Basic construction

The SM-1 is constructed in an integral manner, starting from the T400 tank. Pieces are simply attached to it. No subassemblies are used, but certain parts require precision positioning and reusing already built savefiles is encouraged.

Only a single variant is used.

Guidance system

The stage is usually guided by the gimballing rocket exhaust and whatever method is used in the capsule being carried by it. Since the stage is designed for naked capsule mounting and the only type of the proper size is a Mk-1 pod, the capsule usually provides its gyros to assist in guidance.

The three relatively small fins provide upper stage stability without excessively shifting the centre of pressure when attached to a properly sized lower stage.

Re-entry system

Re-entry is achieved passively. The SM-1 has a centre of mass close to its engine, which allows it to passively obtain the proper re-entry orientation. The fins used for launch stabilisation are destroyed during re-entry and need to be replaced during post-flight maintenance. They are cheap and don't add much to the maintenance time if KCT is used.

Recovery system

SM-1 recovery is achieved via parachutes. The parachute system is overengineered both as a redundancy measure and because of the impossibility to use a single central parachute at its technology level (lacking cubic struts). The service module can land with full propellant tanks. This excessive braking capacity can be used to recover a damaged capsule by landing with the service module attached.

Assembly instructions

Assemble rocket starting from one of the core parts. Store a savefile of the assembled stage for further reference after assembly.

Parts list

- 1x FL-T400 Fuel Tank
- 1x LV-909 "Terrier" Liquid Fuel Engine
- 1x TD-12 Decoupler
- 3x Basic Fin
- 2x Mk2-R Radial-Mount Parachute

Technology

The rocket requires a level 4 technology level.

- Start
 - Basic fins
- Advanced Rocketry
 - Terrier engine
 - T-400 tank
- Engineering 101
 - Decoupler
- Survivability
 - Parachutes

Assembly procedure

- Start with T400 tank
 - Use checkerboard pattern during assembly
- Mount Terrier engine axially at the bottom of the tank
- Mount TD-12 axially on top of tank
- Mount fins radially on tank
 - 3x symmetry with snap
 - W/SE/NE direction
 - As low as possible (centre fin rivet should lay in line at centre of checkerboard)
- Mount Mk12 drogue chutes radially on tank?

- 2x symmetry with snap
 - E/W direction
 - Centre of parachute at lower junction of "rivet ring" panel
- Mount Mk2 parachutes radially on tank?
 - 2x symmetry with snap
 - N/S direction
 - Centre of parachute at lower junction of "rivet ring" panel
- Re-root to decoupler
- Repaint as desired



Figure 1: SM-1 final assembly with marks showing positioning of parachutes and fins

Stage configuration

Part	Variable	Value	Default	Notes
Engine	Thrust limiter	100	Yes	
	Throttle	Main throttle	Yes	
LFOX tank	Flow Priority	0	Yes	
Decoupler	Force percent	100	Yes	

Part	Variable	Value	Default	Notes
Parachutes	Enable crossfeed	False	Yes	
	Enable staging	True	Yes	
	Deploy mode	When safe	Yes	
	Min pressure	0.45	No	
	Altitude	500	No	
	Spread angle	10	No	

Staging

Stage	Parts	Notes
0	Parachutes, decoupler	
1	Engine	

Parameters

Parameter	Value
Mass (dry, kg)	1020
Mass (wet, kg)	3020
Cost (wet)	1965
Cost (dry)	1781
Part count	8

Post flight maintenance

- SM-1 Decoupler
 - Enable staging
- Tanks
 - Fill

Pre flight checks

- Verify staging
- Verify parachute settings